

CLICK DIGITAL SOLUTION

Biratnagar-2, Morang, Nepal

PROJECT DOCUMENTATION

Restaurant Order Calculator

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1. Introduction

This document presents the detailed project documentation for the "Restaurant Order Calculator" a web-based application developed during a self-learning program at Click Digital Solution, Biratnagar. The purpose of this document is to provide a comprehensive overview of the project, including its objectives, design approach, technical implementation, features, and potential areas for future improvement.

The Restaurant Order Calculator is a front-end web application designed to streamline the order management process in a restaurant setting. It allows staff members to efficiently select menu items, calculate order totals, manage delivery charges, and generate order confirmations — all within a clean, responsive interface.

2. Project Overview

The Restaurant Order Calculator is a static, client-side web application that does not require any server infrastructure or backend database. It operates entirely within the user's web browser, making it lightweight, fast, and easy to deploy on any device with a modern browser.

The application serves the following primary purposes:

- Display a restaurant menu with item names and prices
- Allow users to add or remove items from an order using quantity controls
- Calculate and display a live order summary with itemized pricing
- Support multiple order types: Dine In, Takeaway, and Delivery
- Apply automatic delivery charges when the Delivery option is selected
- Collect customer details including name, phone number, and address
- Validate user input before processing the order
- Generate a formatted order confirmation with a unique order number and timestamp

3. Objectives

Primary Objective: To develop a functional, user-friendly web application that simplifies restaurant order processing and calculation.

Skill Development: To strengthen understanding of HTML5, CSS3, and vanilla JavaScript through hands-on project implementation.

Design Philosophy: To create a clean, minimal, and responsive user interface that works seamlessly across desktop and mobile devices.

Practical Application: To build a real-world applicable tool that demonstrates the potential of front-end web development in solving everyday business problems.

4. Technologies Used

The application is built using only core web technologies with no external frameworks or libraries. This approach was chosen to build a strong foundation in fundamental web development concepts.

Technology	Version / Details	Purpose
HTML5	Semantic Markup	Page structure and content layout
CSS3	Flexbox, Grid, Media Queries	Styling, responsive design, and animations
JavaScript	ES6+ (Vanilla)	Application logic, DOM manipulation, event handling
Google Fonts	Inter (300, 400, 500, 600)	Modern, clean typography
CSS :has() Selector	Modern CSS Feature	Radio button active state styling

5. Project Structure

The project follows a simple, flat directory structure with no nested folders. This keeps the project easy to understand and maintain.

File Name	Lines of Code	Description
index.html	117 lines	Main HTML document containing the page structure, sections, and semantic markup
style.css	507 lines	All styling rules including layout, colors, typography, responsive media queries, and transitions
script.js	356 lines	Complete application logic including menu rendering, order management, validation, and order confirmation generation

6. Features and Functionality

6.1 Dynamic Menu Display

The restaurant menu is defined as a JavaScript array of objects, each containing the item name and price. On page load, the menu items are dynamically rendered into the DOM. The current menu includes

five items: Chicken Burger (Rs. 300), Veg Burger (Rs. 270), Chicken Pizza (Rs. 500), Chicken Biryani (Rs. 400), and French Fries (Rs. 170).

6.2 Quantity Controls

Each menu item features increment (+) and decrement (-) buttons that allow the user to adjust the quantity of each item in the order. The quantity cannot go below zero, ensuring data integrity throughout the ordering process.

6.3 Live Order Summary

As items are added or removed, the Summary section updates in real time. It displays each ordered item with its name, quantity, unit price, and line total. The subtotal is automatically recalculated with every change.

6.4 Order Type Selection

The application supports three order types: Dine In, Takeaway, and Delivery. Users can select their preferred option using radio buttons. The selected option is visually highlighted with an inverted black and white style for clear indication.

6.5 Automatic Delivery Charge

When the Delivery option is selected, a delivery charge of Rs. 100 is automatically added to the order total. Additionally, a delivery address input field appears, prompting the user to enter the delivery location. Switching to Dine In or Takeaway removes the charge and hides the address field.

6.6 Customer Information Collection

The application collects essential customer details including full name, phone number, and address. These fields are required before the order can be placed.

6.7 Form Validation

Before placing an order, the application performs comprehensive validation. It checks that at least one item is in the order, the customer name is provided, the phone number is filled, the address is entered, and (if applicable) the delivery address is specified. Missing information triggers browser alert dialogs and focuses the cursor on the offending field.

6.8 Order Confirmation

After successful validation, a detailed order confirmation is generated and displayed. This includes a unique order number (ORD- followed by the last six digits of the current timestamp), a date and time stamp, customer details, order type, itemized receipt with subtotals, and the grand total.

6.9 Responsive Design

The layout is fully responsive and mobile-friendly. The main container has a maximum width of 520px, and a CSS media query at 480px switches the radio button group to a vertical layout and wraps menu items to ensure readability on smaller screens.

6.10 Clean, Minimal UI

The interface follows a card-based design with subtle box shadows, a monochrome color palette (black, white, and gray), smooth hover and active transitions on buttons, and clean typography using the Inter font family.

7. User Interface Design

The user interface is designed with simplicity and usability as top priorities. The layout consists of four main sections arranged vertically within a centered container:

- Menu Section: Displays all available food items with their prices and quantity controls (+/- buttons).
- Summary Section: Shows a live-updating list of ordered items with quantities, prices, and the running subtotal.
- Order Type Section: Contains three radio buttons for selecting Dine In, Takeaway, or Delivery.
- Customer Section: Provides input fields for the customer's name, phone number, address, and (conditionally) delivery address.

The design employs a monochrome color scheme with black buttons, white backgrounds, and gray accents. Card-based sections with subtle shadows create visual depth. Transitions and hover effects provide feedback during user interaction, enhancing the overall user experience.

8. Code Overview

8.1 HTML Structure

The HTML file (Index.html) establishes the page structure using semantic elements. It includes a main container div that houses four primary sections: Menu, Summary, Order Type, and Customer. Each section is clearly labeled and structured for both visual clarity and accessibility.

8.2 CSS Styling

The CSS file (style.css) handles all visual presentation. Key aspects include a flexbox-based layout, responsive design with media queries for screens under 480px, smooth transitions on interactive elements, a monochrome color palette, card-based section styling with subtle shadows, and the use of the modern CSS `has()` selector for radio button state management.

8.3 JavaScript Logic

The JavaScript file (script.js) contains all application logic. It manages the menu data, renders menu items dynamically, handles quantity adjustments, updates the order summary in real time, calculates totals including conditional delivery charges, validates form inputs, and generates the order confirmation with a unique order number and timestamp. The code is structured using modern ES6+ syntax with clean function organization.

9. How to Run the Application

As a static front-end application, no server, database, or special runtime environment is required. The application can be run by following these steps:

Step 1: Locate the project folder: Restaurant_Calculator

Step 2: Open the Index.html file in any modern web browser (Google Chrome, Mozilla Firefox, Microsoft Edge, etc.)

Step 3: The application will load immediately and be ready for use.

Alternative Method:

Open a terminal or command prompt, navigate to the project directory, and run the command: start Index.html (Windows). The application will open in the default web browser.

10. Testing and Validation

The application has been tested for the following scenarios to ensure reliability and correctness:

- **Menu Display:** All five menu items load correctly with accurate prices.
- **Quantity Adjustment:** Increment and decrement buttons function properly. Quantities do not go below zero.
- **Order Summary:** Items appear in the summary only when quantity is greater than zero. Subtotal updates correctly.
- **Delivery Charge:** Rs. 100 is added only when Delivery is selected and removed when Dine In or Takeaway is chosen.
- **Form Validation:** Alerts appear when required fields are left empty. Focus shifts to the invalid field.
- **Order Confirmation:** A unique order number is generated. All details display correctly including timestamp.
- **Responsive Layout:** The interface adjusts properly on screens smaller than 480px.
- **Cross-Browser Compatibility:** The application works correctly on Chrome, Firefox, and Edge.

11. Challenges and Solutions

Dynamic Menu Rendering

Challenge: Rendering menu items dynamically from a JavaScript array required careful DOM manipulation.

Solution: Used template literals and `forEach` loops to create and append HTML elements for each menu item.

Real-Time Summary Updates

Challenge: Keeping the order summary synchronized with quantity changes required efficient event handling.

Solution: Attached event listeners to quantity buttons that trigger a summary update function on every click.

Conditional Delivery Logic

Challenge: The delivery address field and charge needed to appear/disappear based on order type selection.

Solution: Used change event listeners on radio buttons to toggle visibility of the address field and adjust totals.

Responsive Design

Challenge: Ensuring the layout remained usable on very small screens without breaking the design.

Solution: Implemented CSS media queries at 480px to adjust layout properties such as radio group orientation.

12. Future Enhancements

While the current version of the application is fully functional, several enhancements could be implemented to expand its capabilities:

- Backend integration with a server-side framework (e.g., Node.js, Django) for data persistence
- Database connectivity to store orders, menu items, and customer information
- Admin panel for managing the menu (adding, editing, or removing items)
- Payment integration with eSewa, Khalti, or other local payment gateways
- Order history and analytics dashboard for business insights
- User authentication system for staff access control
- Print-friendly receipt generation
- Multi-language support for broader accessibility

13. Conclusion

The Restaurant Order Calculator project successfully demonstrates the practical application of front-end web development technologies in solving a real-world business problem. Through this project, a solid understanding of HTML5, CSS3, and vanilla JavaScript has been developed, including concepts such as DOM manipulation, event handling, responsive design, and form validation.

The application is lightweight, fast, and requires no external dependencies, making it highly portable and easy to deploy. The clean, intuitive interface ensures that it can be used by restaurant staff with minimal training. While there is room for future enhancements, the current version meets all the core requirements of a basic restaurant order management tool.

This project serves as a strong foundation for further learning and development in web technologies, and it demonstrates the ability to translate theoretical knowledge into a functional, user-oriented application.

14. References

- MDN Web Docs — <https://developer.mozilla.org>
- W3Schools — <https://www.w3schools.com>
- Google Fonts (Inter) — <https://fonts.google.com>
- CSS-Tricks — <https://css-tricks.com>
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